

Cyclical Ground Bounce Demo PCB

1. PCB Overview

This PCB is designed to demonstrate the commonly encountered phenomena in high speed digital design known as Ground Bounce. It demonstrates this effect by using a 6 input D-Flip Flop that charges capacitors at Q0-Q4, then simultaneously dumping the capacitors into the ground plane. It repeats this as the clock is fed through an RC delay circuit as the master reset signal. This spec sheet provides an overview of the board design and its use.

Item	Note
Board Size	34mm x 44mm (rectangular)
Stack-Up	2-Layer FR4, 1oz copper
Bottom Layer	Solid Ground Plane
Top Layer	Local Vcc pours(just used vias), Signals

2. Component List & Placement Details

Designator	Part/Package	Purpose	Placement Notes
U1	CD74HCT174PW	6 bit D-FF IC	Center of top layer for shortest q-via paths

C1-C5	0402 (100pF, 75pF, 47 pF, 33pF, 20 pF)	Load Capacitors for Q0-Q4	Bottom layer, directly below Q pins
R2	0402 1k Ω	Q5 Pull down	Next to U1 pin 15
R1/C6	0402 1k Ω + 1nF	RC delay for reset	C6 next to pin 1, R1 on bottom layer under U1
C7	0402 100nF	VCC decoupling cap	Close t o VCC pin, through to ground plane
J1/J2	2 pin headers	Clock in, VCC in	Bottom Edge of board
J3	SMA Edge Jack	Coax probe at q5	Top edge of board

Note: A linear voltage regulator was mistakenly not included in our design, meaning the ground bounce measured at the COAX cable is not seamlessly adjustable, although still visible.

3. Electrical Specifications

Parameter	Typical
Supply Voltage (via 2 pin header)	+5.0V
Quiescent I_{CC}	$\approx 5\text{mA}$
Transient I_{CC} (all Q low)	$\approx 35\text{mA}$ peak
Clock Input	100kHz Square, 5Vpp, 50% duty cycle
Reset Pulse Width (τ)	1ms
Expected Bounce Spike	$\approx 40\text{mV}/2\text{ns}$ at coax cable

4. Ground Bounce Measurement Technique

-Set DC power supply to +5V, and attach to the J2 VCC pin headers at the bottom left of the board.

-Set waveform generator to a 100kHz and 5Vpp square wave with a 50% duty cycle to act as out clock.

-Attach oscilloscope to the coax jack at Q5, ensure 50 Ω input impedance.

-The circuit should already be functioning! The oscilloscope should be showing a repeating spike in voltage relative to ground at the coax output. This spike repeats with respect to the time constant, τ , of our RC delay circuit.

Expected Output:

Note: Unfortunately due to the board delays we decided to move forward with the project using a bread board version of our circuit. The circuit did function as we intended, giving us the output seen in figure 4-1. Checking the clock and reset pins with a probe we also confirmed that they were both getting a fairly clean square wave, with the master reset getting the signal delayed by the RC circuit.



Figure 4-1: Ground Bounce Spike (measured at COAX)

A clear spike in voltage seen at ground.



Figure 4-2: Repeating Ground Bounce Spikes

This shows our RC delay is working as intended.

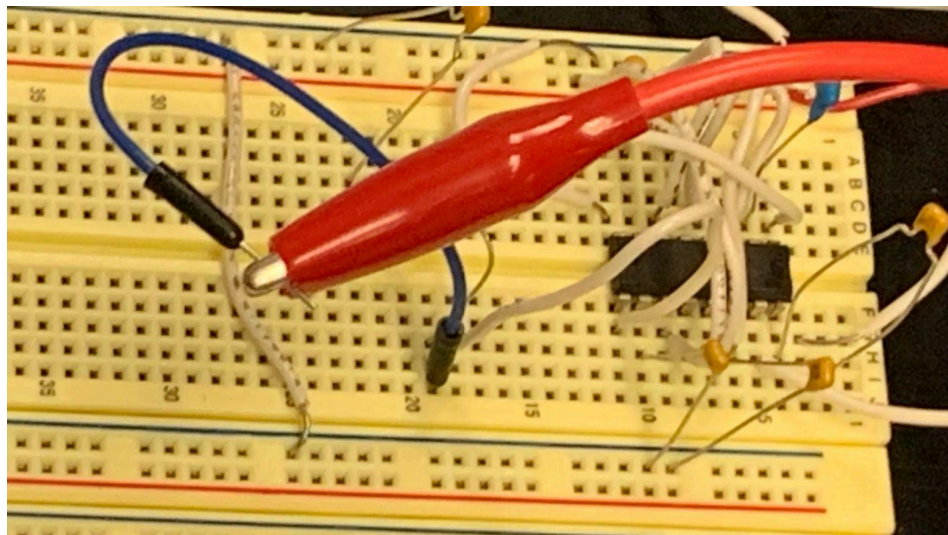


Figure 4-3: Breadboard Circuit

Our breadboard version of the circuit.

5. Schematic and Layout

Ground Bounce Demo Board

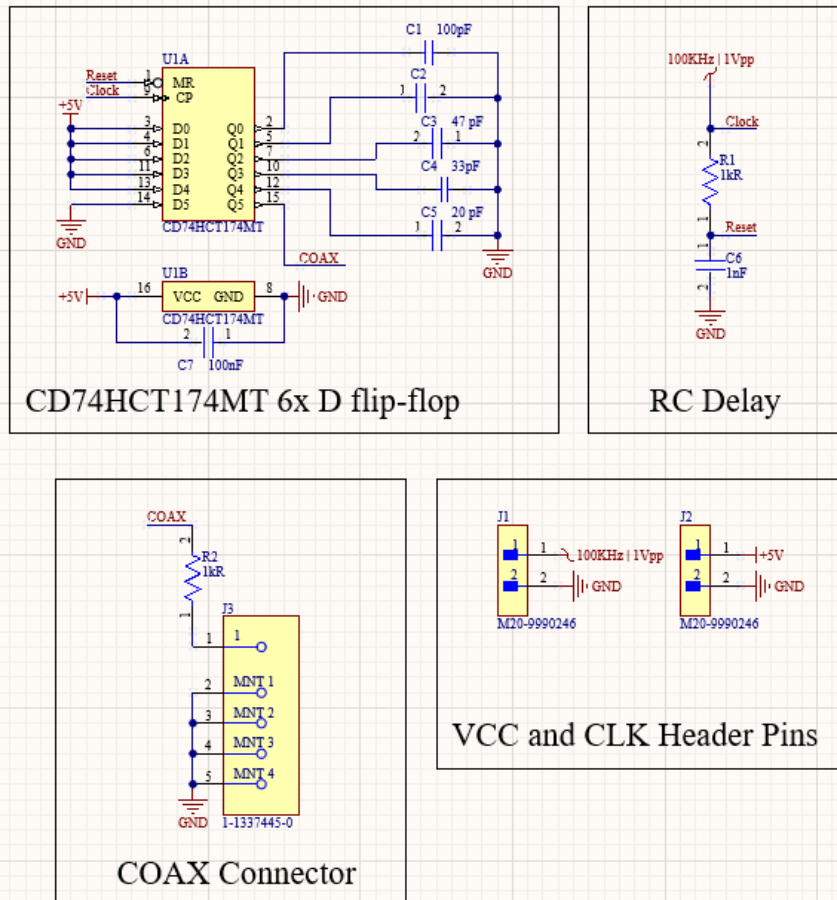


Figure 5-1: Schematic

Design includes a decoupling capacitor between the VCC and GND pins of U1 to limit noise. The RC delay introduces a delay with a time constant of 1ms, allowing time for the capacitors at Q0-Q4 to charge up before the clock signal reaches the master reset pin.

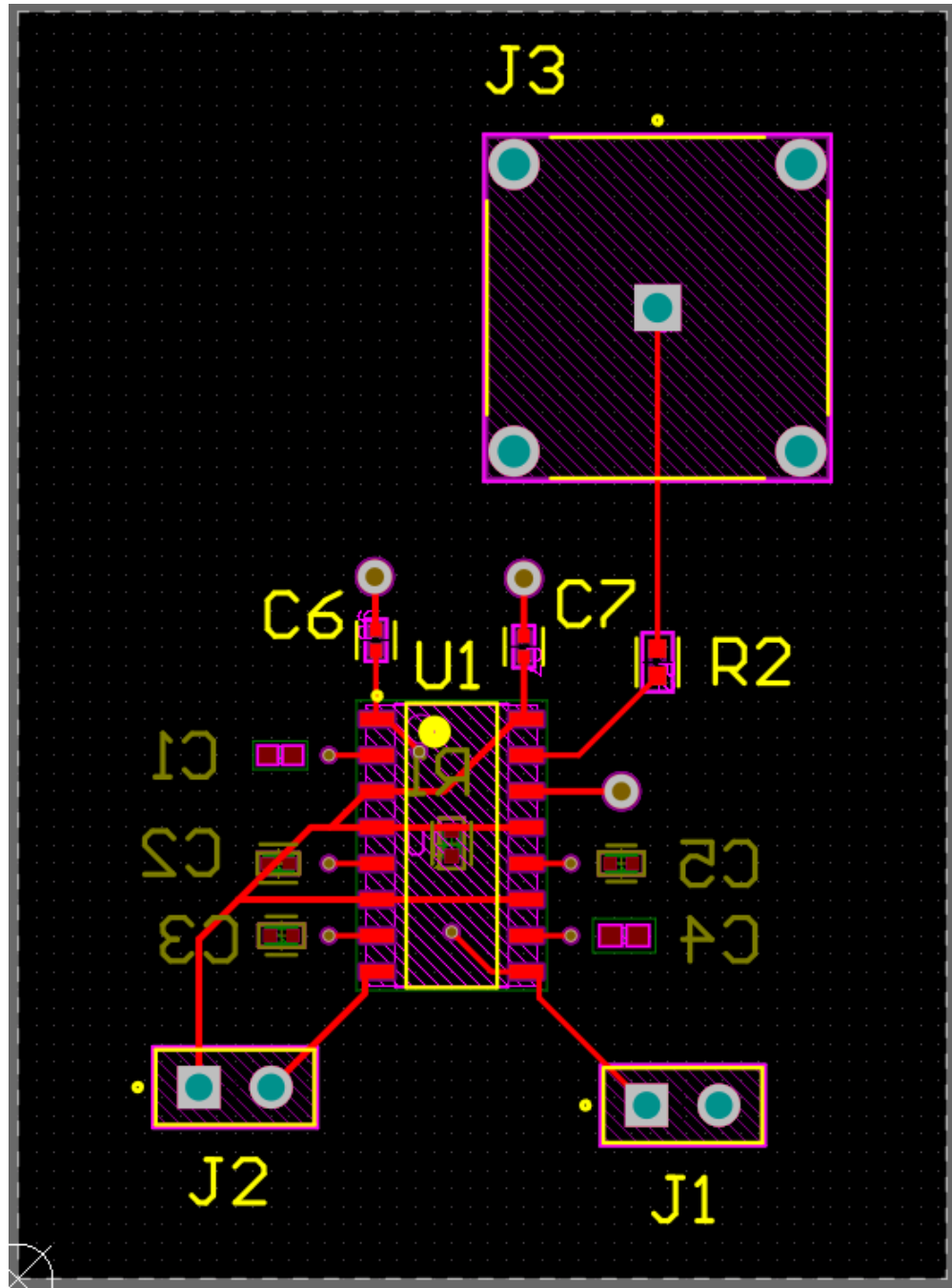


Figure 5-2: PCB Layout (top layer)

This Layer Contains the main flip flop IC itself. The headers were placed at the bottom of the board to make the board feel a bit more organized while keeping trace length low.

Placing C1-C5 on the bottom layer allowed for easier access for VCC and the clock to the pins that needed them.

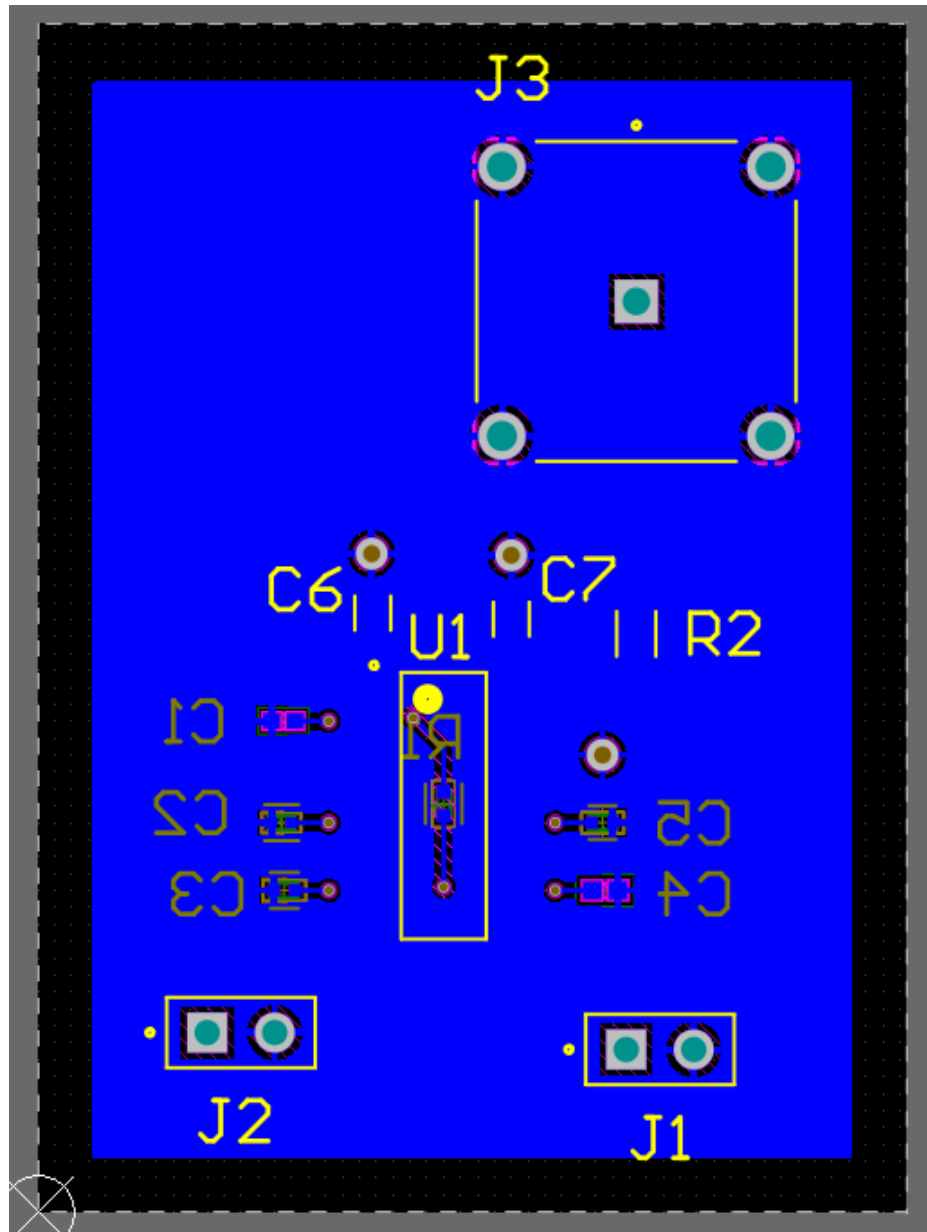


Figure 5-3: PCB Layout (bottom layer)

This layer contains a ground plane for the board, as well as the capacitors for Q0-Q4 and the resistor for the RC delay circuit.

6. Project Questions

a) Bondout Inductance

Can be calculated using the formula $L = \frac{V_{bounce}}{\frac{di}{dt}}$.

Our measured $V_{bounce} \approx 40\text{mV}$.

$\frac{di}{dt} = \text{slew rate} \approx 15\text{mA/ns}$

$L \approx 2.67\text{nH}$

This is quite a bit lower than the typical 5-25nH

b) Edge speed before integrity failure

The faster the edge the higher the slew rate, which increases inductive voltage drops on Ground and VCC corresponding to the equation seen in question a. The typical rise/fall times of this family of ICs is ~2-6ns. Any edges faster than 2ns is where signal integrity issues would begin to appear. A higher $\frac{di}{dt}$ from edge speeds 1ns or faster are highly likely to see ground bounce.

c) Other Bondout Inductances

The other IC pins do have bondout inductance, of them the most likely to require consideration are the pins at VCC, CLK/RST and I/O pins.

Rapid current draw when switching can cause a voltage dip in VCC which could reduce logic high signals. At the I/O pins fast transitions can cause ringing or crosstalk, and is even more important if the pin is

driving a long trace. At the clock and reset pins delays may be introduced, and it could lead to setup and hold time violations

d) Separate Analog and Digital Grounds

Digital switching can cause noisy currents in ground, these fast signals cause a high $\frac{di}{dt}$ which leads to things like ground bounce. Analog circuits rely on a stable ground to reference for accuracy, meaning if they shared a ground the analog signals could be corrupted by the noise created by the digital signals. Keeping separate grounds in a mixed signal system can help to prevent digital signals effects on ground from interfering with the integrity of the analog signals.

e) Multiple Power and Ground pins

Using multiple pins for power or ground can help improve bondout inductance. Having more pins means more parallel paths for the signal to travel through leading to lower total inductance, in turn reducing ground bounce and voltage drops. In higher speed digital ICs pins can draw significantly more current, having multiple paths for the current to flow allows a higher total I_{CC} without exceeding any one pins limits. It can also allow for greater flexibility in the layout.

7. Group Contribution Statement

Member	Tasks Contributed To
Cameron Dubois	PCB Layout, Routing, Gerber generation/submission
Calvin He	Schematic, RC delay calculations, finding correct parts in manufacturer part search
Collaboration	Breadboard building, ground bounce observation, troubleshooting, any in lab work

Note: I would argue that neither one of us did any one chunk of the project in its entirety on our own. There was constant collaboration and checking info or work with the other.